HUANG Yuntong

Computational materials science • phase-field modeling • microstructure evolution

yt.huang@connect.ust.hk
https://github.com/Hatomzre
+852 6467 6876



Education

2023 – present **Ph.D. in Mathematics**, Hong Kong University of Science and Technology.
Research focuses on phase-field modeling and semiphysical modeling of polycrystalline materials. Expected completion: August 2027.

2020 – 2023 **M.Sc. in Computational Mathematics**, Wuhan University.
Thesis: *Optimization based on energy-splitting methods for two-dimensional discrete dislocation dynamics*.

2016 – 2020 **B.Sc. in Materials Science and Engineering**, Central South University.
Thesis: *Numerical simulation of aging precipitation kinetics in Al-Zn-Mg-Cu aluminum alloys*.

Selected Research

2026 – present **VN-GraphGG: graph-based grain growth modeling**.
Developed a two-dimensional grain-growth framework combining von Neumann–Mullins kinetics with graph neural networks for microstructure reconstruction.
Preprint: arXiv:2610.13418.

2023 – 2025 **Deformation-induced amorphization in materials**.
Developed a thermodynamically consistent phase-field model coupling elastoplasticity and stress-dependent transformation strain to investigate amorphization as a deformation mechanism.

2025 – present **Misorientation-dependent grain-boundary grooving**.
Formulated a Kobayashi–Warren–Carter (KWC) phase-field model coupling conserved density, crystallinity, and orientation to study misorientation-dependent grain-boundary grooving.
Preprint: SSRN:7593495.

Research Publications and Preprints

1. **Y. Huang**, S. Dai, C. Chen, and Y. Xiang, "Investigating amorphization as a deformation mechanism using a novel phase field model at the mesoscale," *International Journal of Solids and Structures*, vol. 324, p. 113 691, 2026. DOI: 10.1016/j.ijsolstr.2025.113691
2. **Y. Huang**, Z. Li, S. Dai, C. Ma, and Y. Xiang, *VN-GraphGG: Von Neumann–Mullins–based graph grain growth simulation*, Preprint, Oct. 9, 2026. arXiv: 2610.13418 [cond-mat.mtrl-sci]. URL: https://arxiv.org/abs/2610.13418
3. **Y. Huang** and Y. Xiang, *Phase-field modeling of grain-boundary grooving with misorientation-dependent energy*, SSRN preprint, Oct. 10, 2026. DOI: 10.2139/ssrn.7593495
4. N. Yan, **Y. Huang**, and Y. Xiang. "OmniRemesh: Adaptive and quasi-differentiable remeshing for crystal plasticity simulation and inverse parameter calibration under large deformation." arXiv: 2608.15842 [cs.CE]. URL: https://arxiv.org/abs/2608.15842
5. **Y. Huang** and S. Dai. "New methods derived from energy minimization problems for solving two dimensional discrete dislocation dynamics." arXiv: 2211.15831 [cond-mat.mtrl-sci]. URL: https://arxiv.org/abs/2211.15831

6 Q. Zhang, **Y. Huang**, X. Chen, A. Pan, Z. Cai, S. Liu, and Y. Zhang, "CoF2 nanoparticles grown on carbon fiber cloth as conversion reaction cathode for lithium-ion batteries," *Journal of Alloys and Compounds*, vol. 805, pp. 539–544, 2019. DOI: 10.1016/j.jallcom.2019.07.034

Miscellaneous Experience

Awards

2025 **HKUST RedBird Academic Excellence Award**, Hong Kong University of Science and Technology.
2023 **Hong Kong PhD Fellowship**, Hong Kong Research Grants Council.
2018 **National Scholarship**, Ministry of Education of the People's Republic of China.

Academic Service

2026 Organizer, **EASIAM 2026 Special Student Session**.
2025 Hosted the **HKUST SIAM Student Chapter Annual Meeting 2025** in Hong Kong.

Selected Presentations

2026 **EASIAM 2026 Special Student Session**, Jeju, South Korea.
Presentation: *Phase-field modeling of grain-boundary grooving with tunable misorientation-dependent energy*.
International Conference on Plasticity, Damage, and Fracture, Geelong, Australia.
Presentation: *A multi-scale phase-field model for amorphization as a deformation mechanism in nanocrystalline materials*.
2024 **16th World Congress on Computational Mechanics**, Vancouver, Canada.
Presentation: *A numerical study of shear-induced amorphization in alloy*.
17th East Asia Section of SIAM Annual Conference, Macau SAR, China.
Presentation: *A phase-field model for deformation-induced amorphization*.

Skills

Programming — Python, MATLAB, C++, and LaTeX.
Scientific computing — Phase-field modeling, variational methods, numerical PDEs, elastoplasticity, discrete dislocation dynamics, and microstructure simulation.
Machine learning — Graph neural networks and neural modeling.
Language — English (IELTS 7.0).